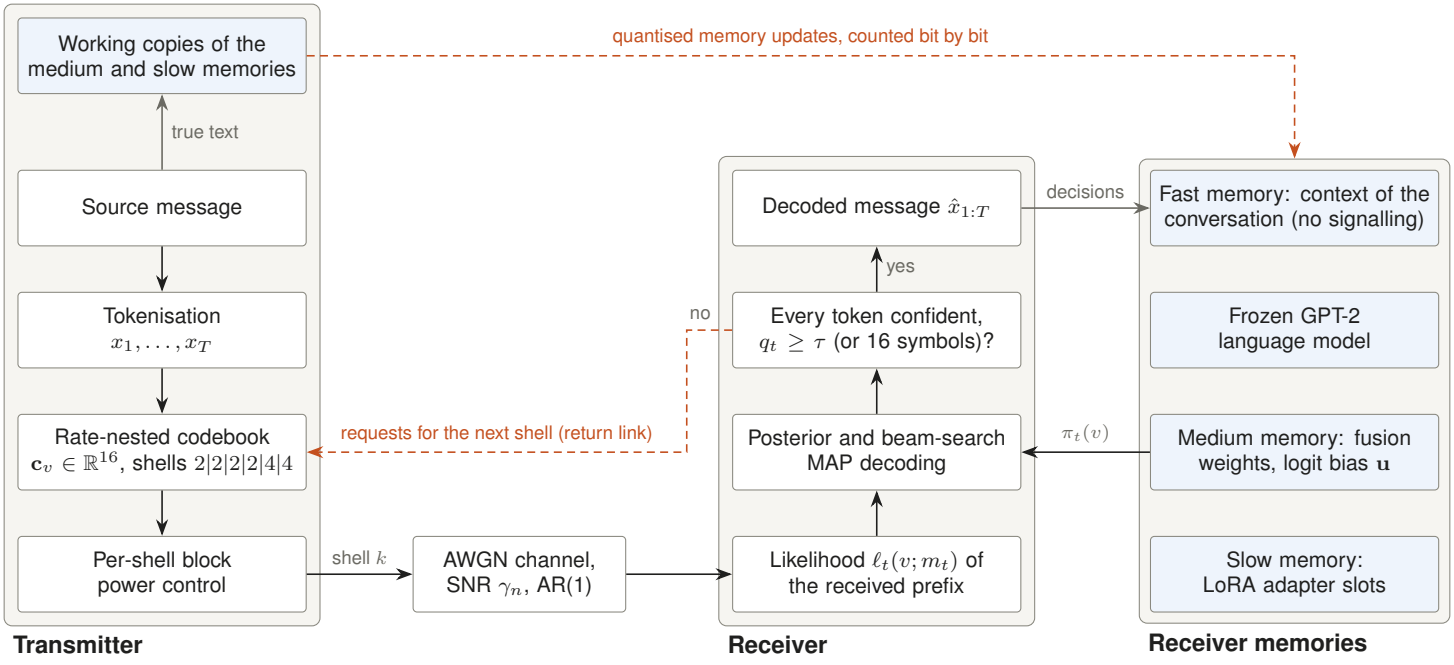

Working copies of the medium and slow memories
quantised memory updates, counted bit by bit
true text
Source message
Tokenisation
$x_1, \ldots, x_T$
Rate-nested codebook
$\mathbf{c}_v \in \mathbb{R}^{16}$, shells $2|2|2|2|4|4$
requests for the next shell (return link)
Per-shell block power control
shell $k$
AWGN channel, SNR $\gamma_n$, AR(1)
Likelihood $\ell_t(v; m_t)$ of the received prefix
Posterior and beam-search MAP decoding
$\pi_t(v)$
Every token confident, $q_t \geq \tau$ (or 16 symbols)?
no
yes
Decoded message $\hat{x}_{1:T}$
decisions
Fast memory: context of the conversation (no signalling)
Frozen GPT-2 language model
Medium memory: fusion weights, logit bias $\mathbf{u}$
Slow memory: LoRA adapter slots
Transmitter
Receiver
Receiver memories